JEE Platform — Question Tagging Specification

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This document lists every field that is tagged on a problem in the `jee_platform` question bank. Fields marked **[NEW]** were added by the Stage 1 PRD; everything else was locked in `docs/PROJECT CONTEXT.md`. The total surface for a fully-tagged problem is approximately 35 fields, split across the 11 groups below.

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1 The 7 problem-identity axes (existing)

These describe what *kind* of problem this is. All seven are required at insert time and indexed individually. They form the question code: `TOPIC.SUBTOPIC.IDEA.SUBIDEA.NNN`.

1.1 `TOPIC` — broad subject family (~12 values)

Code	What it tags
ALG	Algebra (quadratic, polynomial, complex numbers, binomial)
PNC	Permutations & Combinations
PRB	Probability
MAT	Matrices & Determinants
TRG	Trigonometry
CAL	Calculus
COG	Coordinate Geometry (2D)
VEC	Vectors & 3D Geometry
SET	Sets, Relations, Functions
SOT	Sequences & Series
LOG	Logarithms (often subsumed under ALG)

1.2 `SUBTOPIC` — chapter section under the topic (~5–8 per topic)

Listed per topic in `/content/taxonomy/maths.yaml`. Examples:

- For `MAT`: `ALG` (matrix algebra), `DET` (determinants), `INV` (inverse/adjoint), `SYS` (linear systems), `SPL` (special: symmetric/orthogonal/permutation), `TRN` (transformations/conjugation).
- For `CAL`: `FUN`, `LIM`, `CON`, `DIF`, `DER`, `IND`, `DEF`, `ARE`, `ODE`.
- For `PNC`: `DGT` (digit-based counting), `PER`, `COM`, `DST`, `INC`, `REC`, `BIJ`.

1.3 `IDEA` — the principle being tested (the MOST important axis)

The smallest insight such that — with it, the problem is routine; without it, impossible regardless of computation skill. Per (TOPIC, SUBTOPIC). Grows organically. Examples: `ORBSUM` (sum over a group orbit gives an orbit-invariant), `EXMUL` (exact-distinct multiplicity counting).

1.4 `SUB-IDEA` — the specific manoeuvre under the idea ($\leq$ 6 per idea)

The exact mechanical trick. Per (TOPIC, SUBTOPIC, IDEA). Examples: `CNJSP` (conjugation by a permutation permutes row + col simultaneously), `LZINC` (leading-zero inclusion-exclusion correction).

1.5 `ANSWER-TYPE` — the format of the answer (5 closed values)

Code	Meaning
MCQ-SC	Multiple choice, single correct
MCQ-MC	Multiple choice, multi-correct (with partial marking)
NUM-INT	Numerical answer, integer
NUM-DEC	Numerical answer, decimal (to specified precision)
MAT-COL	Match-the-column / match-the-list

1.6 `SURFACE` — cognitive-load dressing (NOT the math) (6 closed values)

Code	Meaning
SURF-PLAIN	Plain statement, minimal dressing
SURF-SET	Heavy set / index / summation notation
SURF-FUNC	Wrapped in function definitions
SURF-GEOM	Geometric figure or diagram
SURF-PARAM	Parameterized — letters as the real unknowns
SURF-PASS	Comprehension passage (long preamble)

1.7 `TRAP` — the bait the problem dangles (8 closed values)

Code	Meaning
TRAP-NONE	No specific trap
TRAP-EIGEN	Tempts eigenvalue / spectral reasoning
TRAP-CAYLEY	Tempts Cayley–Hamilton invocation
TRAP-LHOP	Tempts L'Hôpital's rule where elementary manipulation works
TRAP-NCERT	Tests an NCERT edge case
TRAP-EDGE	Tests an edge case in domain / range / continuity
TRAP-PARTIAL	Multi-correct calibrated for partial-marking play
TRAP-LENGTH	Long-looking problem with short solution (or vice versa)

2 The 5 diagnostic / failure-mode axes [NEW]

These describe what *kind of student error* the problem activates. Tagged on each `wrong_paths` entry, NOT on the problem row directly. 19 axis values total.

2.1 `ERR-READING` — misread the statement (4 values)

Value	When to tag
ERR-READING-NONE	No reading mistake on this path.
ERR-READING-QUANTIFIER	Misread a quantifier (“exactly”, “at least”, “contains”, “from each”, “non-zero”, “all”).
ERR-READING-NUMRANGE	Misread a numeric bound or interval endpoint.
ERR-READING-VARDEF	Misread a variable’s defining property or domain.

2.2 `ERR-CASE` — case-handling failure (4 values)

Value	When to tag
ERR-CASE-NONE	No case-handling mistake.
ERR-CASE-EDGE	Boundary or degenerate case dropped or double-counted (leading-zero, $x = 0$, equality of two parameters).
ERR-CASE-MODINDEX	Wrong period / parity / cycle length on an iterated structure (recurrence, matrix power, modular).
ERR-CASE-PARTITION	Missing or duplicate sub-case in a case enumeration.

2.3 `ERR-COMP` — computation slip (4 values)

Value	When to tag
ERR-COMP-NONE	No computation slip.
ERR-COMP-ARITH	Arithmetic slip (a number added / multiplied wrong).
ERR-COMP-SIGN	Sign or orientation flipped (vector direction, negative of an integral, wrong root sign).
ERR-COMP-ALGEBRA	Algebraic manipulation slip (wrong identity applied, factoring error).

2.4 `ERR-STRATEGY` — strategic / meta error (3 values)

Value	When to tag
ERR-STRAT-NONE	No strategy error.
ERR-STRAT-TRAP	Took the bait — invoked trap machinery (eigen / Cayley / L'Hôpital / NCERT-edge / length-of-problem) when a simpler method exists.
ERR-STRAT-PARTIAL	Multi-correct partial-marking miscalibration (picked too many or too few options for the marking scheme's expected-value math).

2.5 `ERR-PARSING` — failure to map statement to math (4 values)

Value	When to tag
ERR-PARSE-NONE	Statement was parsed correctly.
ERR-PARSE-LISTMATCH	Wrong mapping in List-I / List-II match (each computation correct in isolation).
ERR-PARSE-GEOM3D	Failed to form / hold the geometric or 3D figure correctly.
ERR-PARSE-PASSAGE	Misunderstood the comprehension passage / multi-paragraph setup.

Special pattern: "didn't see the IDEA"

The all-`NONE` pattern across the 5 diagnostic axes signals a *pure conceptual failure* on the existing IDEA / SUB-IDEA axes (Group 1). This is the only failure type the original 7-axis model isolates. The platform displays it as: *"didn't see the IDEA: [idea_label]."*

3 Difficulty fields

3.1 `authored_difficulty` — T1–T5 intrinsic rating (existing)

Fixed forever, anchored to a top-10-rank student in their last 4–5 months of preparation.

Code	Meaning
T1	Easy standard
T2	Standard medium
T3	Hard standard
T4	Hard non-standard
T5	Top-100 only

3.2 `empirical_difficulty_by_round` — nullable, computed from real attempts

JSON object, one entry per round (R1–R4). Computed by the nightly Python batch job once a problem has ≥ 30 attempts. Each value is the success rate among students at or above the problem's intended round. Null while `status = provisional`.

4 Time fields

4.1 `authored_time_by_round` — per-round expected time, in seconds (existing)

JSON object with 4 keys (`R1_seconds`, `R2_seconds`, `R3_seconds`, `R4_seconds`). The *hypothesis* for how long a top-10-rank student takes per round. Example from `MAT.SPL.ORBSUM.CNJSP.001`: $\{R_1 : 780, R_2 : 540, R_3 : 360, R_4 : 210\}$ seconds.

4.2 `empirical_time_by_round` — per-round actual median, in seconds (nullable)

JSON object, same shape as authored. Computed by the nightly batch job once ≥ 30 attempts exist. Each value is the *median* time among students who answered correctly. The authored-vs-empirical gap is itself useful signal (it measures and improves authoring intuition).

5 The R1–R4 round model

A per-student, per-fingerprint attribute — **NOT an 8th axis of problem identity**. Stored on the `student_fingerprint_state` table.

Code	Stage of preparation
R1	First Prep — learning the idea; near-zero mastery.
R2	First Revision — "seen it" → "can do it reliably."
R3	Second Revision — integration; two-concept fusion, speed, surface variation.
R4	Final Round — optimisation; exam simulation, strategy, retention.

Four levers change per round (the problem itself stays the same): inclusion (which problems to show), time expectation (same problem, solved faster each round), expected role (teaches idea → speed → triage), pass/fail threshold (rises each round).

6 Identifier fields

6.1 `question_code` — natural primary key

Unique string of the form `TOPIC.SUBTOPIC.IDEA.SUB-IDEA.NNN`, e.g. `MAT.SPL.ORBSUM.CNJSP.001`. Used as foreign key from `tests` (which store ordered arrays of codes) and `attempts` (one row per student-per-question-per-attempt).

6.2 `serial` — the NNN sequence within the fingerprint

Integer ≥ 1. Sequence within (TOPIC, SUBTOPIC, IDEA, SUB-IDEA). Together with the 4 fingerprint codes it forms the natural uniqueness constraint.

7 Content fields

7.1 `statement` — the question text

Markdown + LaTeX (KaTeX / MathJax-compatible). Rendered to the student. The full surface they see.

7.2 `answer` — the correct answer (shape depends on `ANSWER-TYPE`)

- For `MCQ-SC` / `MCQ-MC`: `{type, correct_options: ["A", "B", "C"]}`.
- For `NUM-INT` / `NUM-DEC`: `{type, value: <number>, precision: <int>}` (the `precision` field is being added in Stage 2 to support the equality contract for matching wrong paths).
- For `MAT-COL`: `{type, list_i_to_list_ii: {1: "P", 2: "Q,R", ...}}`.

7.3 `solution` — the canonical worked-out solution

Markdown + LaTeX. Must use JEE-syllabus methods only (no eigenvalue decomposition for matrix problems unless the syllabus permits, etc.). Verified to produce the answer in `answer`.

7.4 `wrong_paths` — array of plausible student-mistake records

JSON array. Typically 2–4 entries per problem. Each entry is a structured sub-object (see Group 8 below). Provides the diagnostic signal for the 5 new failure-mode axes.

8 Per-wrong-path subfields

Each entry in the `wrong_paths` array is itself a structured record with these fields:

8.1 `path` — natural-language description of the mistake

What the student did wrong, in 1–3 sentences. Example: *"Forgets that the diagonal frequency depends on n. Uses '2 times' (from the $n = 3$ case in JEE 2019 P2 Q1) instead of '6 times' for $n = 4$."*

8.2 `landed_on_option` — the wrong answer this mistake produces

For MCQ: the letter(s), e.g. `"D"` or `"A and B"`. For NUM: the value the wrong path produces, e.g. `"`β `= 52"`. This is what the wrong-path matcher uses to identify which path a real student took.

8.3 `diagnosis` — short why-they-failed explanation

1–2 sentence explanation of the underlying error. Example: *"Failure to generalize the orbit-frequency counting from S_3 to S_n."*

8.4 `diagnostic_tags` [NEW] — the 5 axes from Group 2

Object with exactly 5 keys, one value per axis from Group 2:

- `err_reading`: one of `ERR-READING-{NONE, QUANTIFIER, NUMRANGE, VARDEF}`
- `err_case`: one of `ERR-CASE-{NONE, EDGE, MODINDEX, PARTITION}`
- `err_comp`: one of `ERR-COMP-{NONE, ARITH, SIGN, ALGEBRA}`
- `err_strategy`: one of `ERR-STRAT-{NONE, TRAP, PARTIAL}`
- `err_parsing`: one of `ERR-PARSE-{NONE, LISTMATCH, GEOM3D, PASSAGE}`

So tagging one wrong path = 5 picks, one per axis. Tagging a problem with 3 wrong paths = 15 picks total. This is the calibration tagging burden.

9 Provenance & lifecycle

9.1 `status` — where the problem is in its lifecycle (2 values)

Value	Meaning
provisional	No human approval yet AND/OR no empirical data ($<$ 30 attempts). Mutable.
calibrated	Human-approved AND has $\geq$ 30 empirical attempts. Immutable — importer refuses to overwrite.

9.2 `source_metadata` — provenance object

JSON object. Sub-fields:

- `type`: `generated` (Claude wrote it) or `refined` (existed somewhere, polished here).
- `inspired_by`: the source identifier of the JEE Advanced question or coaching problem that anchored this one. E.g. `jeea-2019-p2-q1` or `case-based-questions-2-q1`.
- `parallel_form`: boolean. True if axes 1–4 (TOPIC, SUBTOPIC, IDEA, SUB-IDEA) match the inspiration — a parallel-form mastery-check rather than variety practice.
- `generator`: `claude`, `<human-author-name>`, or other.
- `created_on`: ISO date.
- `human_reviewer`: who approved this problem (null until approved).
- `approved_at`: ISO date of human approval (null until approved).

9.3 Timestamps — `created_at`, `updated_at`

DB-managed. `created_at` is set on first insert; `updated_at` auto-updates on every row change.

10 Derived / summary fields [NEW]

At the problem level, computed deterministically from the `wrong_paths` array. The Stage 2 Architect picks the mechanism (Postgres generated column / trigger / view). Once landed, application code MUST NOT be able to write to them — drift between summary and source is structurally impossible.

10.1 `failure_modes_seen` — distinct non-NONE values used (array)

Array of strings. For each of the 5 diagnostic axes, the distinct non-`NONE` values that appear across the problem's wrong paths. GIN-indexed for fast queries like *"find all problems where `ERR-PARSE-GEOM3D` fires."*

10.2 Other summary columns proposed by the Architect

The Stage 2 Architect may propose additional summary columns (e.g. per-axis activation count). All must satisfy the same drift-proof invariant.

11 Per-attempt fields (live on `attempts`, not `problems`)

When a student makes an attempt on a problem, a row is appended to the `attempts` table (append-only — never UPDATE'd or DELETE'd). The attempt references the problem by `question_code`, and carries these fields:

- `student_id` — FK to `students`.
- `question_code` — FK to `problems`.
- `test_id` — FK to `tests`; nullable (null means a standalone practice attempt).
- `correct` — boolean, did they get the right answer.
- `time_seconds` — how long they spent on this question, measured silently by the test runtime.
- `visit_count` — how many times they returned to this question (1 = single visit; 2+ = navigated away and came back).
- `marked_for_review` — boolean, did they tick the "mark for review" button.
- `attempt_order` — 1st, 2nd, 3rd attempt by this student at this question across all time.
- `round_at_time` — which round (R1–R4) the student was in when they attempted this. Per the round model in Group 5.
- `hints_used` — integer count of hints / tips revealed during the attempt.
- `created_at` — timestamp.

These fields feed the nightly batch job that computes the *empirical* versions of difficulty and time (Groups 3.2 and 4.2). They also drive the wrong-path matcher: given a student's wrong answer + their `landed_on_option`, the system looks up the matching `wrong_paths` entry to identify which failure modes activated.

Summary — total tagging surface for one problem

Group	Fields	Notes
1. Identity axes	7	All required, enums or strings
2. Diagnostic axes **[NEW]**	5 per wrong-path	15 picks if 3 wrong paths
3 + 4. Difficulty + Time	4	2 authored, 2 empirical (nullable)
6. Identifier	2	question_code + serial
7. Content	4	Includes wrong_paths array
8. Per-wrong-path subfields	4 per entry	Including diagnostic_tags
9. Provenance	8 sub-fields	status + source_metadata + timestamps
10. Derived / summary **[NEW]**	≥ 1	Auto-computed
11. Per-attempt fields	11 (on attempts)	Not on problems table
Total fields on a fully tagged problem	**∼35**	Most required at insert

Net new burden added by Stage 1 PRD: the 5 `diagnostic_tags` values per wrong-path entry. For a typical 3-wrong-path problem: 15 extra picks. At ∼4 min/problem steady state (∼8–12 min during the first 30 questions of calibration warm-up). Total upfront calibration cost: ∼15 reviewer-hours.

Source-of-truth files: `agent-factory/scorecards/01-prd-final.md` *for the Stage 1 PRD;* `docs/PROJECT CONTEXT.md` *for the binding 7-axis spec;* `content/taxonomy/maths.yaml` *for the working enum value catalogue.*